MATERIAL	SPEC NUMBER	REVISION	EFFECTIVE DATE
AISI 4340, 125K, Low Alloy Steel NON-NACE	CMS-118	B	12/17/2024

Material Requirement

Scope	AISI 4340, 125K Minimum Yield Strength, Low Alloy Steel NON-NACE
Product Forms	Bar and Mill Shapes, Ring Forgings, Open and Closed Die Forgings
General Requirements	CMS-002, Material Requirements for Forgings, Bars and Mill Shapes

Chemical Composition

Element	Composition Limits	Element	Composition Limits
Carbon	0.38 – 0.43	Silicon	0.15 - 0.35
Manganese	0.60 – 0.80	Chromium	0.70 – 0.90
Phosphorous	0.025 Maximum	Molybdenum	0.20 - 0.30
Sulfur	0.025 Maximum	Nickel	1.65 – 2.00
Vanadium	–	Iron	Balance

Mechanical Properties

Properties	Range
Tensile Strength, psi	140,000 (965 MPa) Minimum
Yield Strength, psi, 0.2% Offset	125,000 (862 MPa) Minimum
Elongation in 2" Gage Length	16% Minimum
Reduction in Area	32% Minimum
Brinell Hardness (kg/sq mm)	31 – 35 HRC(293 – 327 HBW)

Melt Practice	Electric furnace process with subsequent vacuum treatment (EFVD). Vacuum induction melting (VIM), vacuum arc remelting (VAR), or Electroslag remelting (ESR)
Condition	Material shall have an ASTM grain size of #5 or finer and a minimum hot work ratio of 4:1.
Heat Treatment General	Material shall be provided in the normalized (optional), quenched and tempered condition.

Heat Treatment Normalize

Atmos/Media	Temperature	Time at Temp
Air	1600°F - 1700°F (871°C - 927°C)	½ Hr/Inch of Maximum Thru Thickness. 1 Hour Min

Still Air Cool to Below 400°F (204°C) before processing to the next stage.

Heat Treatment Austenitize

Atmos/Media	Temperature	Time at Temp
Air	1550°F - 1625°F (843°C - 885°C)	½ Hr/Inch of Maximum Thru Thickness. 1 Hour Min

Temperature shall be at least 25°F less than normalizing temperature

Heat Treatment Quench

Atmos/Media	*Time/Time at Temperature*
Water	100°F (38°C) Maximum Before Quenching 120°F (49°C) Maximum at any time during the quench cycle
Polymer or Oil	Per the quenchant manufacturer's standards *Start temperature for polymer or oil quenchant shall be per the quenchant manufacturer.*

Document the start and finish temps for quench media.

	Atmos/Media	Temperature	Time at Temp
Heat Treatment Temper	Air	1150°F - 1300°F (621°C - 704°C)	3/4 Hr/Inch of Maximum Thru Thickness 1 Hour Min
	Air cool or faster to room temperature.		

Heat Treatment Continuous Furnace

For bars with diameters of 8" (203 mm) or less, continuous furnace heat treatment is an acceptable alternative to conventional batch-type heat treatment.

	Parameters	Time at Temp
Austenitizing Furnace	Bar temp exiting final zone	1550°F - 1625°F (843°C - 885°C)
Austenitizing Furnace	Min time	5 minutes (Note)
Water Spray Quench		
Tempering Furnace	Bar temp exiting final zone	1150°F - 1300°F (621°C - 704°C)
Tempering Furnace	Min time	5 minutes (Note)
Air cool or faster to room temperature		

- *The bar travels through several different temperature zones which allows the bar to heat to an austenitic state.*
- *Prior to entering the tempering furnace, the bar is spray quenched. Max water temperature at start of quench is 100°F(38°C).*
- *The tempering furnace zone temps are then selected to produce the required tempering effect.*
- *The length of the furnace and travel speed determined the time spent in the austenitizing and tempering furnace and the bar diameter affects the travel speed.*
- *To attain the required mechanical properties and microstructure, the time in each furnace should produce the same mechanical properties as a conventional quench-and-temper heat treatment.*

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Date Prepared	*December 17, 2024*	***Date Approved:***	*December 17, 2024*

Revision History Table

Revision	Date	Description
B	12/17/2024	Reduced limit of Phosphorous and Sulfur to .025 max.
A	07/29/2021	Initial release